The Great Blue Hole

A stunningly beautiful submerged sinkhole in a reef off the coast of Belize

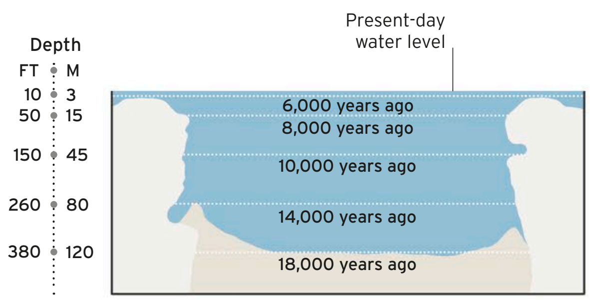
The Great Blue Hole is situated on Lighthouse Reef, an atoll lying 35 miles (55 km) east of the huge Belize Barrier Reef. It is a large, almost perfectly circular sinkhole within the limestone mass of the reef.

Formation and exploration

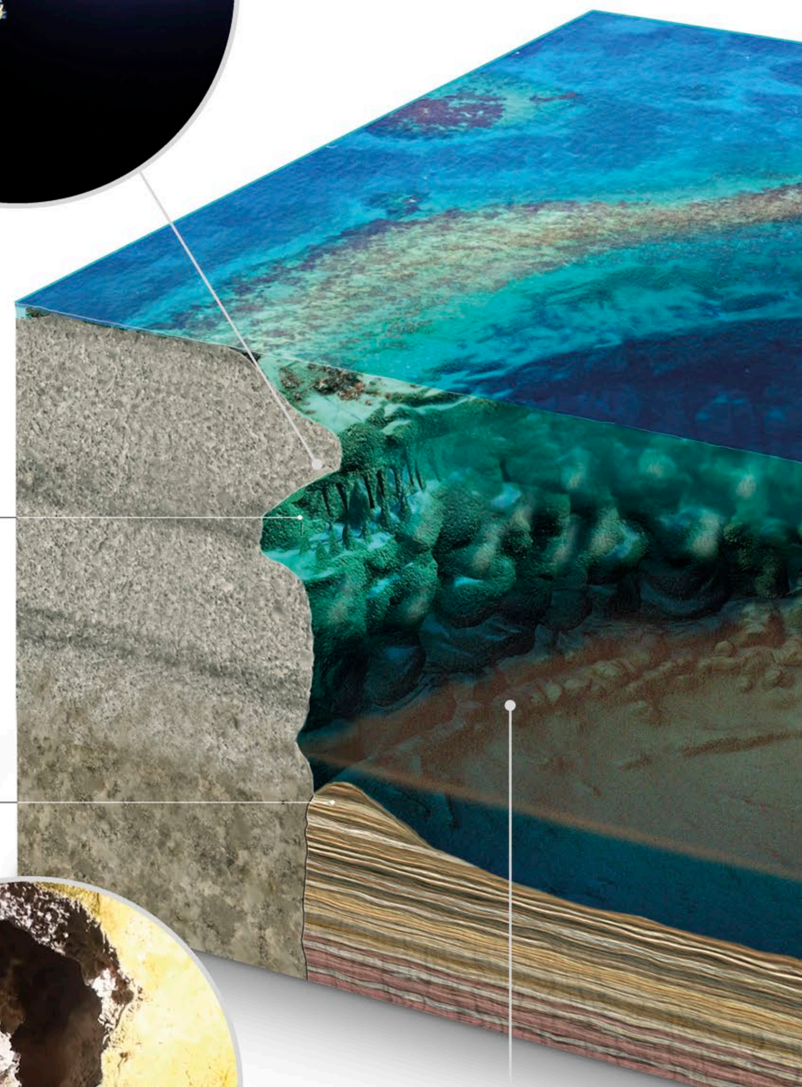
Analysis of stalactites found in the Great Blue Hole shows that it formed in stages going back at least 153,000 years, during periods when sea level was lower than it is today. Erosion by streams during these times produced a complex of air-filled caves and tunnels in a block of limestone on dry land. Eventually, the ceiling of one cave collapsed, creating what is now the entrance to the Great Blue Hole. Since its formation, sea level has gone up and down, most recently rising to flood the blue hole and a cave system linked to it. Today, both the sinkhole and caves are accessible only to adventurous scuba divers but are regarded as one of the world's most exciting dive sites. Diving into the hole is not recommended for novice divers because perfect buoyancy control is required. Caribbean reef sharks are often seen swimming around the entrance.

PREVIOUS SEA LEVELS

The Great Blue Hole has not always been underwater. It formed over several tens of thousands of years through erosion of a mass of limestone on dry land. The sea level has risen over the past 18,000 years, since the last ice age, causing the Great Blue Hole to become submerged.



◁ **STALACTITE EVIDENCE**
The presence of stalactites on several locations around the walls of the hole are proof that it formed above sea level, since stalactites and stalagmites cannot form underwater.

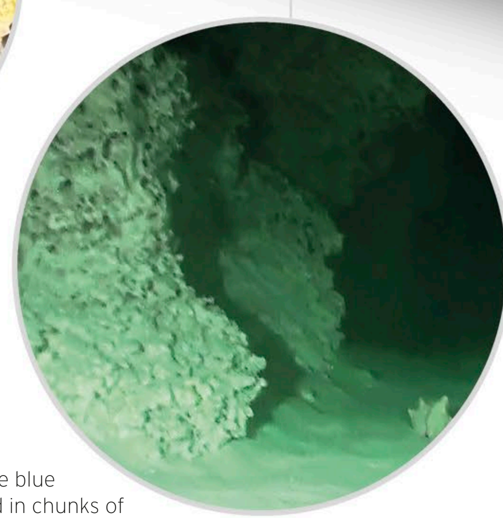


South grotto cave
Large stalactites hang from the cave roof and stalagmites project from its floor, giving the entrance the appearance of an open shark's mouth.

Sediment dune
This is created by coral fragments raining down from the lip of the shelf some 360 ft (110 m)

△ **WALL CAVITY**
The sinkhole's walls are pockmarked with small cavities. These provide ideal hiding places for some marine animals, except at great depth, where there is a lack of oxygen in the water.

▲ **THE GREAT BLUE HOLE**
With a diameter of 1,043 ft (318 m), the Great Blue Hole is 407 ft (124 m) deep and has rough-textured limestone walls indented by ledges, grottoes, and caves. Its entrance is almost completely surrounded by a ring of reef projecting slightly above the sea surface.



▷ **AT THE BOTTOM**
The floor of the blue hole is covered in chunks of limestone and fallen, broken stalactites, as first reported by the underwater adventurer Jacques Cousteau, who used a small submersible to explore the sinkhole in 1971.